6	(a)	The isotope beryllium-7 $\left({}^{7}_{4}\mathrm{Be}\right)$ is radioactive with a half-life of 53 days. A beryllium-7 nucleus decays by the emission of a γ-ray photon of energy 0.48 MeV.		
		(i)	Calculate the change in mass of a beryllium-7 nucleus when it undergoes γ-decay.	
			change in mass = ……kg	[2]
		L1	Recap: • Mass-energy of the nucleus is converted into electromagnetic radiation energy (photon). • Thus, mass of nucleus decreases after it emits a photon. • Nucleus is heavier in the excited state. Using Einstein's mass-energy equivalence $E = mc^2$: (Energy of photon emitted) = (change in mass) c^2 **$0.48 \times 10^6 \times 1.60 \times 10^{-19}$ = (change in mass) $(3.00 \times 10^8)^2$** **M1** (change in mass) = 8.5333×10^{-31} = **8.5×10^{-31} kg (or 8.53×10^{-31} kg)** **A1** ***Marks scheme:*** ***Wrong conversion of MeV to J is penalized under A1. Award M1 for correct application of (Energy of photon emitted) = (change in mass) c^2***	
		(ii)	Determine the probability of decay of a beryllium-7 nucleus in a time of 1.0 day.	
			probability = ……	[2]
		L2	Recap: The **decay constant** is defined as the **probability** of a radioactive nucleus decay in the **next unit time interval**. probability of decay in one day	

			= the decay constant with a unit of day^{-1} = $\frac{\ln 2}{53 \text{ days}}$ = 0.013078 = **0.013** day^{-1} **(or 0.0131** day^{-1}**)** ***Do not penalise if unit is not stated in the final answer.***	**M1** **A1**
		(iii)	Initially, the total mass of undecayed beryllium-7 in a radioactive sample is 5.7×10^{-12} kg. Calculate the number of undecayed beryllium-7 nuclei in the sample after 120 days.	
			number = ……	[3]
		L2	number of undecayed nuclei $N = N_o e^{-\lambda t}$ where N_o = number of undecayed nuclei at time t = 0 = initial mass of sample / mass of a nucleus = $\frac{5.7 \times 10^{-12} \text{ kg}}{7 \text{ u}}$ = $\frac{5.7 \times 10^{-12} \text{ kg}}{7 \times 1.66 \times 10^{-27} \text{ kg}}$ number of undecayed nuclei $N = N_o e^{-\lambda t}$ = $\frac{5.7 \times 10^{-12} \text{ kg}}{7 \times 1.66 \times 10^{-27} \text{ kg}} e^{(-0.013078 \times 120)}$ = 1.0212 x 10^{14} = **1.0 x 10^{14} (or = 1.02 x 10^{14})** ***Marks scheme:*** ***Wrong / Lack of conversion of u to kg is penalized under A1.*** ***Award 1st M1 for evidence of knowledge that 7u is the nuclear mass.*** ***Award 2nd M1 for correct application of*** $N = N_o e^{-\lambda t}$***.***	**M1** **M1** **A1**
		(iv)	Explain why, although the beryllium-7 is radioactive, the number of beryllium-7 nuclei in the sample does not change.	
			……	
			……	

			.....	
			.....	[2]
		L2	A **γ-ray photon is a quantum of** (electromagnetic radiation) **energy** (and has no proton and neutron). **When a Be-7 nucleus emits a γ-ray photon, its proton number and neutron number are unchanged. The type of nuclide thus remains unchanged. (OR, The type of nucleus depends on the proton number and the neutron number in the nucleus. Thus the Be-7 nucleus remains as Be-7 nucleus after the emission of a γ-ray photon.)**	**B1** **B1**
	(b)	The beryllium-7 nucleus in **(a)** may be considered to be stationary before the emission of the γ-ray photon. Determine the recoil speed of a beryllium-7 nucleus due to the emission of a γ-ray photon.		
			speed = ...m s^{-1}	[3]
		L2	From de Broglie's relation, $p = \frac{h}{\lambda}$ Energy of a photon, $\mathbf{E = hf = \frac{hc}{\lambda} = \left(\frac{h}{\lambda}\right)c = pc}$ ∴ Momentum of a photon $p = \frac{E}{c}$ $= \frac{\mathbf{0.48 \times 10^{6} \times 1.60 \times 10^{-19}}}{\mathbf{3.00 \times 10^{8}}}$ $= 2.56 \times 10^{-22}$ kg m s^{-1} Initial total momentum = 0 since the Be-7 nucleus is at rest initially Final total momentum = (momentum of Be-7) + (momentum of photon)	**M1**

		Take direction of photon's momentum to be positive direction. Let v be the recoil speed of Be-7. By conservation of total linear momentum, Final total momentum = Initial total momentum (momentum of Be-7) + (momentum of photon) = 0 **$(-7u.v) + (+2.56 \times 10^{-22}) = 0$** **$7\ (1.66 \times 10^{-27})\ v = 2.56 \times 10^{-22}$** **$v = 22000\ m\ s^{-1}$** ***Marks scheme:*** ***Wrong conversion of MeV to J is penalized under A1. [Allow ECF from (a)(i) for wrong conversion of MeV to J.]*** ***Wrong / Lack of conversion of u to kg is penalized under A1.*** ***Award 1st M1 mark for correct values substituted albeit wrong units conversion.*** ***Award 2nd M1 mark for correct application of conservation of momentum with sign convention considered.***	**M1** **A1**

[Total: 12]